FIM- and L2-Based `constraintL2()` Regularization

Current calculation, as implemented

Reference: `scripts/fim_l2_constraintL2.R`, Part 8 (lines 302–342)

This note writes down, step by step, the calculation that currently produces `obj_fim` and `obj_l2` in `fim_l2_constraintL2.R`. It is a transcription of the code, not a proposal for changing it; every quantity below is annotated with the exact line(s) that compute it.

1 Setup and notation

Let $p \in \mathbb{R}^n$ denote the outer (log-parametrized) parameter vector, $n = 28$ in this model. Let

$$\text{best} \in \mathbb{R}^n$$

be the parameter vector returned by the *single-start* `trust()` fit of Part 5 (`line 258--264`: `myfit <- trust(obj, pouter_init, ...); best <- myfit$argument`). Note that this is *not* the later multistart winner `best_ms` (`line 288`) — everything in Part 8 is centered on and evaluated at this earlier, single-fit `best`.

Let obj be the unregularized objective `obj <- normL2(data, g*x*p)` (`line 251`): the weighted least-squares negative log-likelihood of the simulated data under the model.

The reference curvature H

$$H \;=\; \nabla_p^2 \,\mathrm{obj}(p)\big|_{p=\text{best}} \qquad (\texttt{lines 267--268: o_best <- obj(best); H <- o_best\$hessian}) \tag{1}$$

Because `obj(best)` is called with the default `deriv2 = FALSE`, H is *not* the exact second derivative of the objective. It is the Gauss–Newton approximation

$$H \;=\; S^\top W S, \qquad S = \frac{\partial(\text{observable})}{\partial p}\Big|_{p=\text{best}} \text{ (data rows only)}, \qquad W = \mathrm{diag}\big(1/\sigma_{\text{data}}^2\big)\,, \tag{2}$$

i.e. exactly the classical Fisher information matrix built from the model sensitivities and the assumed measurement noise $\sigma_{\text{data}} = 0.015$ (`line 245`). The exact Hessian would add a residual-curvature correction term that only appears when `deriv2 = TRUE` is requested; that term is not part of the current calculation.

2 Eigendecomposition and calibration constants

$$H = V\,\mathrm{diag}(s)\,V^\top, \qquad \texttt{line 310: eig <- eigen(H, symmetric = TRUE); s <- eig\$values} \tag{3}$$

where V's columns are the orthonormal eigenvectors (the model's stiff/sloppy directions) and $s = (s_1, \ldots, s_n)$ the corresponding curvatures.

A numerical-zero mask identifies the practically non-identifiable directions (`lines 270--271`, computed earlier against the same H):

$$\texttt{zero_tol} = 10^{-6} \cdot \max_i |s_i|, \qquad \texttt{is_zero}_i = \mathbb{1}[\,|s_i| < \texttt{zero_tol}\,]\,. \tag{4}$$

Two calibration constants are derived from the spectrum (`lines 312--313`):

$$\texttt{eps_reg} = \min\left\{\, |s_i| : \texttt{is_zero}_i = 0 \,\right\} \quad \text{(smallest strictly-nonzero curvature; a floor)} \tag{5}$$

$$\texttt{strength} = 0.1 \quad \text{(fixed scalar, currently hard-coded)} \tag{6}$$

3 FIM-anisotropic precision → `sigma_fim`

A per-eigendirection weight is assigned, *inverted* relative to the likelihood's own curvature — small s_i (sloppy) gets a large weight, large s_i (stiff) gets a small weight (`line 318`):

$$w_i^{\text{fim}} \;=\; \frac{\texttt{strength}}{s_i + \texttt{eps_reg}}, \qquad i = 1, \ldots, n. \tag{7}$$

These weights are rotated back into parameter space to form a full, generally non-diagonal precision matrix (`lines 319--320`):

$$\text{Prec}_{\text{fim}} \;=\; V\,\text{diag}(w^{\text{fim}})\,V^\top, \qquad \text{symmetrized: } \text{Prec}_{\text{fim}} \leftarrow \tfrac{1}{2}\Big(\text{Prec}_{\text{fim}} + \text{Prec}_{\text{fim}}^\top\Big)\,. \tag{8}$$

Bridging to a diagonal penalty

`constraintL2(mu, sigma)` only implements a diagonal penalty $\sum_i (p_i - \mu_i)^2/\sigma_i^2$, so Prec_{fim} cannot be passed in directly. The diagonal of Prec_{fim} is used instead (`lines 328--329`):

$$\texttt{diag_fim}_i \;=\; \big[\text{Prec}_{\text{fim}}\big]_{ii} \;=\; \sum_k V_{ik}^2\, w_k^{\text{fim}}, \qquad \sigma_i^{\text{fim}} \;=\; \frac{1}{\sqrt{\max(\texttt{diag_fim}_i,\ \varepsilon_{\text{mach}})}}. \tag{9}$$

Why the diagonal, specifically: for the full quadratic form $d^\top \text{Prec}_{\text{fim}}\, d$ with $d = p - \text{best}$, the second derivative with respect to a single p_i, holding every other parameter fixed, is exactly $2\,[\text{Prec}_{\text{fim}}]_{ii}$. `constraintL2`'s diagonal penalty gives $2/(\sigma_i^{\text{fim}})^2$ for that same derivative. Setting $\sigma_i^{\text{fim}} = 1/\sqrt{[\text{Prec}_{\text{fim}}]_{ii}}$ therefore matches the FIM penalty's exact *per-parameter* curvature at best — while necessarily discarding the off-diagonal (cross-parameter) terms $[\text{Prec}_{\text{fim}}]_{ij}$, $i \neq j$, that made the original penalty direction-aware *across* parameters, not just within each one.

4 Isotropic L2 precision → `sigma_l2`

The same weight formula is used, but with a single scalar curvature — the median of the spectrum — standing in for the whole eigenstructure (`lines 334--335`):

$$w^{\text{iso}} \;=\; \frac{\texttt{strength}}{\text{median}(s) + \texttt{eps_reg}}, \qquad \sigma_i^{\text{l2}} \;=\; \frac{1}{\sqrt{w^{\text{iso}}}} \quad \text{for every } i = 1, \ldots, n. \tag{10}$$

Because this target precision is already a scalar multiple of the identity, $\text{Prec}_{\text{l2}} = w^{\text{iso}} I$, the diagonal-only `constraintL2` form reproduces it *exactly* (not merely approximately) — there is no off-diagonal structure to lose in the first place.

5 The two regularized objectives

`constraintL2(mu, sigma)` implements the penalty

$$R(p;\ \mu, \sigma)\ =\ \sum_{i=1}^{n} \frac{(p_i - \mu_i)^2}{\sigma_i^2}. \tag{11}$$

The two regularized objectives are then (`lines 341--342`):

$$\text{obj}_{\text{fim}}(p) = \text{obj}(p)\ +\ \sum_{i=1}^{n} \frac{(p_i - \text{best}_i)^2}{(\sigma_i^{\text{fim}})^2} \qquad \texttt{obj_fim <- obj + constraintL2(mu = best, sigma = sigma_fim} \tag{12}$$

$$\text{obj}_{\text{l2}}(p) = \text{obj}(p)\ +\ w^{\text{iso}} \sum_{i=1}^{n} (p_i - \text{best}_i)^2 \quad \texttt{obj_l2 <- obj + constraintL2(mu = best, sigma = sigma_l2)} \tag{13}$$

where `+` on two `objfn`s sums value, gradient, and Hessian (`"+.fn"`, `R/classes.R`).

6 Summary: what each construction preserves and loses

	FIM-anisotropic	**Isotropic L2**
Curvature input	full spectrum $s_1, \ldots, s_n$ and eigenvectors V	one scalar, median(s)
Resulting σ	one per parameter, σ_i^{fim}, direction-informed via V	one shared value σ^{l2} for all n parameters
Relation to Prec_{fim}	exact on the diagonal only; off-diagonal (cross-parameter) coupling dropped	exact – Prec_{l2} was already diagonal ($= w^{\text{iso}} I$)
Effect on stiff directions	weight $w_i^{\text{fim}} \propto 1/s_i$ vanishes as s_i grows – largely untouched	same absolute pull as everywhere else – can distort well-identified parameters
Effect on sloppy directions	weight $w_i^{\text{fim}} \propto$ `strength`/`eps_reg` near the floor – strong, targeted pull	only as strong as the single global w^{iso} allows

Note: `strength = 0.1` is currently a fixed constant applied identically to both constructions, with no calibration procedure implemented in the script.